

**AMMS1623 CALCULUS AND ALGEBRA FORMULAE****Distance Formula**

The **distance** between the points  $A(x_1, y_1)$  and  $B(x_2, y_2)$  in the plane is

$$d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

**Midpoint Formula**

The **midpoint** of the line segment from  $A(x_1, y_1)$  to  $B(x_2, y_2)$  is

$$\left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

**Distance between a Point and a Line**

The **distance**  $d$  from a point  $Q(x_0, y_0)$  to the line  $ax + by + c = 0$  is

$$d = \frac{|a(x_0) + b(y_0) + c|}{\sqrt{a^2 + b^2}}$$

**Equation of a Circle**

An equation of the circle with **centre**  $(h, k)$  and **radius**  $r$  is

$$(x - h)^2 + (y - k)^2 = r^2$$

**Vectors in Two Dimensions**

If  $\mathbf{u} = \langle a_1, a_2 \rangle$  and  $\mathbf{v} = \langle b_1, b_2 \rangle$ , then their **dot product** is

$$\mathbf{u} \cdot \mathbf{v} = a_1 b_1 + a_2 b_2$$

If  $\theta$  is the **angle** between  $\mathbf{u}$  and  $\mathbf{v}$ , then

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta$$

The **angle**  $\theta$  between  $\mathbf{u}$  and  $\mathbf{v}$  satisfies

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|}$$

**Vectors in Three Dimensions**

The **distance** between the points  $P(x_1, y_1, z_1)$  and  $Q(x_2, y_2, z_2)$  is given by the

$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

If  $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$  and  $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$  are vectors in space, then their **dot product** is

$$\mathbf{u} \cdot \mathbf{v} = a_1 b_1 + a_2 b_2 + a_3 b_3$$


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### The Cross Product of Vectors in Space

If  $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$  and  $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$  are vectors in space, then their **cross product** is the vector

$$\mathbf{u} \times \mathbf{v} = (a_2 b_3 - a_3 b_2)\mathbf{i} - (a_1 b_3 - a_3 b_1)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k}$$

We can calculate the **cross product** using **determinant**.

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

### The scalar triple product

The product  $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$  is called the **scalar triple product** of the vectors  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$ . You can check that the scalar triple product can be written as the following determinant:

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

### Equations of Lines and Planes

A **line** passing through the point  $P(x_0, y_0, z_0)$  and parallel to the vector  $\mathbf{v} = \langle a, b, c \rangle$  is described by the **parametric equations**

$$x = x_0 + at, y = y_0 + bt, z = z_0 + ct$$

where  $t$  is any real number.

A **plane** containing the point  $P(x_0, y_0, z_0)$  and having the normal vector  $\mathbf{n} = \langle a, b, c \rangle$  is described by the **plane equation**

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

### The Distance from a Point to a Line

Distance from a Point  $S$  to a Line Through  $P$  Parallel to  $\mathbf{v}$

$$d = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|}$$

### The Distance from a Point to a Plane

If  $P$  is a point on a plane with normal  $\mathbf{n}$ , then the distance from any point  $S$  to the plane is

$$d = \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|$$

where  $\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$  is normal to the plane.

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**AMMS1623 CALCULUS AND ALGEBRA FORMULAE****Product Rule**

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

**Quotient Rule**

$$\frac{d}{dx} \left[ \frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

**The Chain Rule**

To differentiate  $f(g(x))$ , differentiate first the outside function  $f(x)$  and substitute  $g(x)$  for  $x$  in the result. Then, multiply by the derivative of the inside function  $g(x)$ . Symbolically,

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$$

| <b>Differentiation Formula</b>                       | <b>Corresponding Integration Formula</b>           |
|------------------------------------------------------|----------------------------------------------------|
| $\frac{d}{dx}(x^r) = rx^{r-1}$                       | $\int rx^{r-1}dx = x^r + C$ or                     |
| $\frac{d}{dx}\left(\frac{x^{r+1}}{r+1}\right) = x^r$ | $\int x^r dx = \frac{x^{r+1}}{r+1} + C, r \neq -1$ |
| $\frac{d}{dx}(e^x) = e^x$                            | $\int e^x dx = e^x + C$                            |
| $\frac{d}{dx}(\ln  x ) = \frac{1}{x}$                | $\int \frac{1}{x} dx = \ln  x  + C$                |
| $\frac{d}{dx}(\sin x) = \cos x$                      | $\int \cos x dx = \sin x + C$                      |
| $\frac{d}{dx}(\cos x) = -\sin x$                     | $\int \sin x dx = -\cos x + C$                     |
| $\frac{d}{dx}(\tan x) = \sec^2 x$                    | $\int \sec^2 x dx = \tan x + C$                    |

**Integration by Substitution**

$$\int f(g(x))g'(x)dx = F(g(x)) + C$$

**Integration by Parts**

$$\int f(x)g(x)dx = f(x)G(x) - \int f'(x)G(x)dx$$


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### Linear Approximation

$$f(x) \approx L(x) = f(a) + f'(a)(x - a)$$

The mean value of  $f(x)$  over the interval from  $x = a$  to  $x = b$ .

$$\frac{1}{b-a} \int_a^b f(x) dx$$

The volume of the solid of revolution generated by revolving the region under  $y = g(x)$  from  $x = a$  to  $x = b$  about the  $x$ -axis.

$$\int_a^b \pi [g(x)]^2 dx$$

### Quadratic Formula

$$ax^2 + bx + c = 0$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$